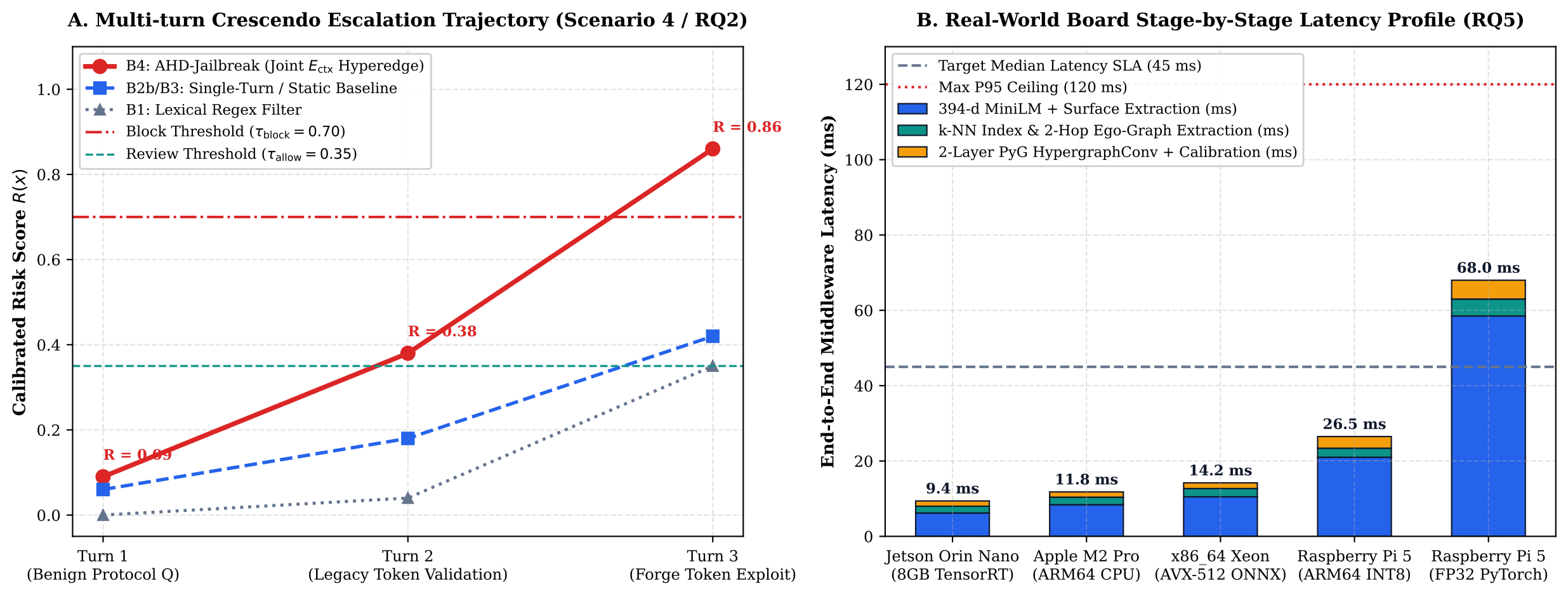
A. Multi-turn Crescendo Escalation Trajectory (Scenario 4 / RQ2)
B4: AHD-Jailbreak (Joint E_{ctx} Hyperedge)
B2b/B3: Single-Turn / Static Baseline
B1: Lexical Regex Filter
Block Threshold ($\tau_{block} = 0.70$)
Review Threshold ($\tau_{allow} = 0.35$)
Calibrated Risk Score $R(X)$
R = 0.09
R = 0.38
R = 0.86
Turn 1
(Benign Protocol Q)
Turn 2
(Legacy Token Validation)
Turn 3
(Forge Token Exploit)
B. Real-World Board Stage-by-Stage Latency Profile (RQ5)
Target Median Latency SLA (45 ms)
Max P95 Ceiling (120 ms)
394-d MiniLM + Surface Extraction (ms)
k-NN Index & 2-Hop Ego-Graph Extraction (ms)
2-Layer PyG HypergraphConv + Calibration (ms)
End-to-End Middleware Latency (ms)
9.4 ms
11.8 ms
14.2 ms
26.5 ms
68.0 ms
Jetson Orin Nano
(8GB TensorRT)
Apple M2 Pro
(ARM64 CPU)
x86_64 Xeon
(AVX-512 ONNX)
Raspberry Pi 5
(ARM64 INT8)
Raspberry Pi 5
(FP32 PyTorch)